



UNIVERSITY OF MINNESOTA

May 4, 2025

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Dear Professor Pope,

I am writing to introduce my formal research report titled *Enhancing Cybersecurity Awareness: Investigating Phishing Preparedness Among UMN Students*. This report focuses on the growing concern of phishing scams and how aware students at the University of Minnesota are when it comes to identifying and responding to threats like these. Throughout this project, I conducted a Google Forms survey that accumulated over 50 responses from students from various departments, as well as interviews with professors and outreach to the university's IT support team.

As someone who has a passion in cybersecurity, I wanted to learn how our campus community responds to phishing attempts, and if more preventative education might help reduce risks. The data suggest that while phishing is commonly encountered, student confidence and behavior vary. This report concludes with recommendations for scalable, student-friendly cybersecurity education options—such as short Canvas modules—to improve awareness and reduce risk across campus.

Thank you for your guidance throughout this course, and the opportunity to study a topic I'm passionate about. I hope you find the report informative and thoughtfully crafted. If you have any questions or concerns, you can reach out to me at tran1385@umn.edu.

Sincerely,

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UNIVERSITY OF MINNESOTA

**Enhancing Cybersecurity Awareness: Investigating Phishing Preparedness
Among UMN Students**

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Cynthia Pope

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Abstract

Phishing attacks are among the most prevalent cyber threats in the current digital age, and college students are frequently targeted. This report aims to address how aware University of Minnesota students are when it comes to phishing, as well as how prepared they feel when facing this issue. I have gathered over 50 responses from Google Forms Surveys, distributed among different channels, such as Canvas Inbox, Reddit, and personal connections. Additionally, I have interviewed faculty members at Carlson School of Management and reached out to the UMN IT department for additional information. The results indicate that most students have encountered phishing emails previously, yet the vast majority do not feel entirely confident in how to identify or handle them. Furthermore, while most students were able to identify typical signs of phishing, a proportion would still open or respond to suspect messages. This research highlights that even though awareness is present among students, there is still a gap in preparedness. As the world continues to rely more on digital environments, this report recommends creating short cybersecurity training tools, such as Canvas modules or phishing simulations, to help students stay informed and make safer decisions online.

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Introduction

Phishing attacks have become a big problem for higher education institutions like the University of Minnesota (UMN). Since students rely on services like email, campus Wi-Fi, and online learning platforms almost every day, it makes them easier targets for cybercriminals (Nyasvisvo & Chigada, 2023). At the same time, colleges store a lot of sensitive information—like student records and research—which makes them even larger targets (Pincus, 2025).

This report investigates how aware UMN students are about phishing attacks, and what can be done to help them recognize and avoid these threats. The main question I’m trying to answer is: *How can the University of Minnesota reduce cybersecurity incidents among students through required awareness training?*

Phishing doesn’t just affect one student—one successful attack could impact many people across campus, and even lead to major problems like stolen data or canceled classes (Pincus, 2025). With cyberattacks getting more advanced every year, it’s important to understand where students stand and what steps the university can take to help protect as many people as possible (Nyasvisvo & Chigada, 2023).

To study this, I created a survey for UMN students and interviewed professors to get a different perspective. I have also contacted the university’s Information Security Office to ask for more information. The results of my survey, the interviews, and research from other studies will be used to better understand the problem and suggest ways UMN can improve student cybersecurity awareness.

Methods

To investigate phishing awareness among UMN students, I collected data using two main methods: a student survey and interviews with faculty members.

First, I created and distributed an online survey using Google Forms. It also required students to use their UMN Internet ID login to ensure that only those within the university can participate, and avoid any potential responses from outsiders that might skew the results. The survey included multiple-choice and short-answer questions aimed at addressing students' knowledge of phishing attacks, their experiences with suspicious emails, and their general cybersecurity habits. It also asked about their confidence in identifying phishing attempts and whether they have fallen for phishing scams, or know someone who has. The survey link was shared through student networks and online platforms. In total, 53 students responded to the survey.

In addition to the survey, I reached out to four UMN professors to gather professional insights and perspectives. Out of the four, two professors provided full interview responses. One professor mentioned that they were not very familiar with the topic and referred me to a colleague, while another initially agreed to participate but later did not respond. The two completed interviews offered perspectives on student cybersecurity behavior, phishing awareness levels, and the importance of education in preventing cyber threats. I also contacted the University of Minnesota Information Security Office to request additional information regarding phishing threats targeting students. A security analyst from the IT department responded, confirming that phishing attempts have increased in recent years and explaining that phishing tactics commonly involve online collaboration documents and fake survey forms. They also

mentioned the availability of Information Security Awareness (ISA) training and ongoing initiatives like MyU tiles and website resources to educate students.

Besides collecting new data, I also reviewed published studies about cybersecurity awareness and phishing in higher education. I found articles through ResearchGate, Google Scholar, and a few cybersecurity training websites. These sources helped in creating survey questions, interview topics, and my overall understanding of the problem.

The combination of student survey results, faculty interviews, and research from academic sources were used for analyzing phishing awareness at UMN and forming recommendations for improvement.

Results

A total of 53 University of Minnesota students completed the phishing awareness survey. Most students reported having seen suspicious messages before, but their confidence in handling phishing attempts varied.

When asked if they had ever received a suspicious email or message, 69.8% of students answered yes, while 28.3% said no, and 1.9% were not sure. According to a study by Nyasvisvo and Chigada (2023), exposure to phishing has increased among students during remote learning, showing that students are frequently targeted by cybercriminals. This supports the idea that a majority of UMN students have already encountered phishing attempts firsthand.

Have you ever received an email or message that you suspected was a phishing attempt?
53 responses

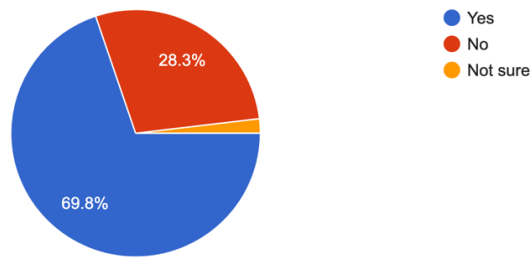


Figure 1 Student responses to whether they have received a suspicious email or phishing attempt.

When it came to feeling confident regarding what to do with a suspicious email/message, 66% said they felt confident, while 30.2% felt somewhat confident, and 3.8% was not confident at all. This shows that even though most students feel prepared, a portion still has doubts. Similar factors were shown by Alharbi and Tassaddiq (2021), who found that even when students recognize threats, they might lack the confidence or training to respond appropriately.

Do you feel confident in knowing what to do if you get a suspicious email/message?
53 responses

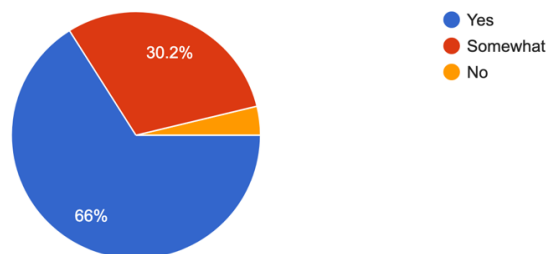


Figure 2 Student confidence in knowing how to respond to a suspicious email or phishing message.

Students were then also asked which signs they would associate with phishing emails. The most common answers were requests for passwords or personal info (90.6%), unfamiliar sender addresses (84.9%), and strange or mismatched links (84.9%). Grammar or spelling errors (75.5%) and urgent or threatening language (81.1%) were also frequently recognized. These insights line up with common phishing tactics described by Rahartomo et al. (2025), who mentioned that phishing often relies on creating urgency and mimicking legitimate-looking messages to trick users.

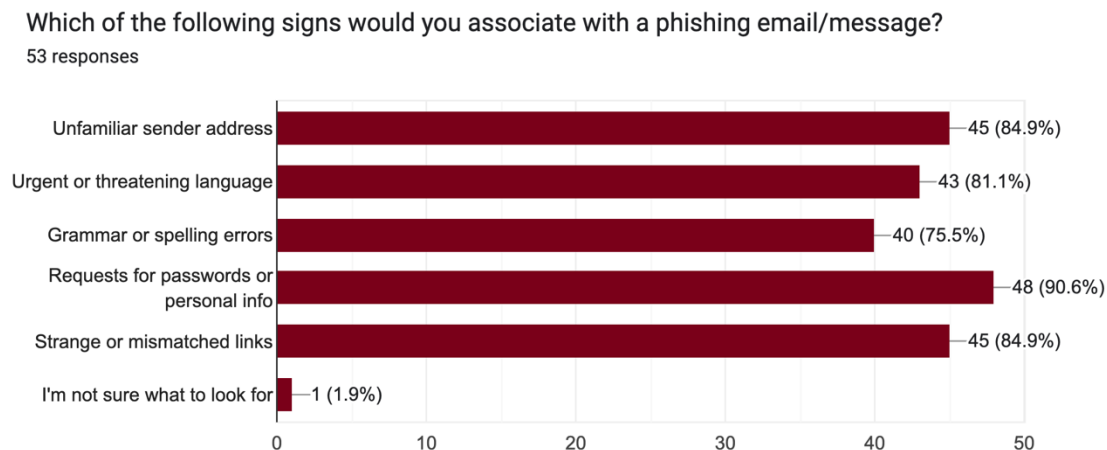


Figure 3 Student recognition of common signs associated with phishing or emails or messages.

As for being asked what they would do after receiving a suspicious email, 67.9% of students said they would delete it immediately, and 60.4% said they would mark it as spam. About 37.7% would report it to the university or IT. A smaller number said they might open it to see what it says (15.1%) or even click the link (3.8%), which matches concerns in the research article about risky student behavior even when warning signs are present (Nyasvisvo & Chigada, 2023).

What would you do if you received a suspicious email/messages?

53 responses

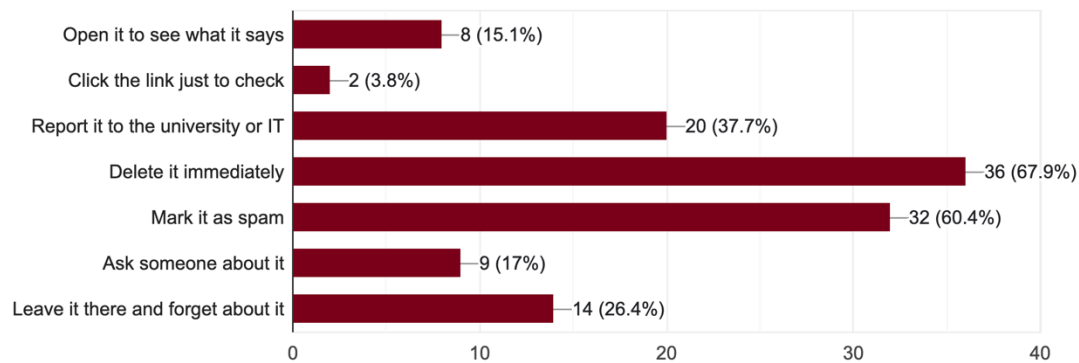
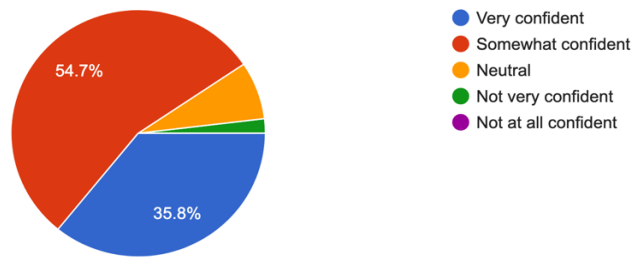


Figure 4 Student actions when receiving a suspicious email or phishing message.

When students rated their confidence in spotting phishing attempts, 35.8% said they were “very confident”, 54.7% were “somewhat confident”, 7.5% felt neutral, and a small percentage selected feeling “not very confident” or “not confident at all”. Overall, most students seemed to have at least some confidence in identifying phishing attacks. However, having confidence does not always mean students will respond correctly. Nyasvisvo and Chigada (2023) found that even students who are familiar with phishing threats can underestimate how convincing some attacks can be. Similarly, Alharbi and Tassaddiq (2021) pointed out that recognizing a phishing sign is one thing, but reacting correctly under pressure is another. Although UMN students show a good level of awareness, these results tell us that there is still room for improvement within the real world of phishing threats.

How confident are you in your ability to identify a phishing attempt?
53 responses



X

Figure 5 Student confidence in their ability to identify phishing attempts

Next, students were asked if they would support a short cybersecurity training module to help prevent phishing attacks. About 26.4% of students said they would definitely support the idea, while 66% said they would support it if it was quick and easy. Only 7.5% said they would not support it. These results show that while not everyone is immediately enthusiastic regarding mandatory cybersecurity training, most students are open to it if the training is made simple and convenient. This goes in line with recommendations from Infosec (2025), which emphasized that cybersecurity education is most effective when it is short, practical, and easy for students to fit into their schedules.

Would you support the implementation of a short cybersecurity training module for all UMN students to help prevent phishing attacks and increase awareness?
53 responses

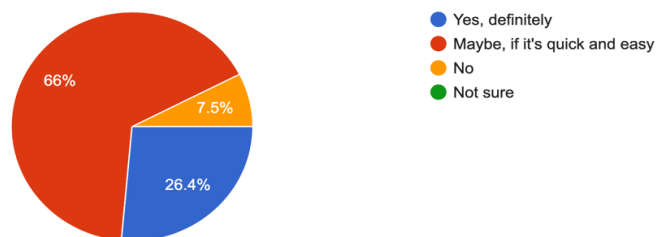


Figure 6 Student support for implementing a short cybersecurity training module at UMN

Finally, an optional question asked students whether they had personally fallen for a phishing scam or knew someone who had. Out of the 21 responses, 38% of students said having no personal experience with phishing, while 29% said they knew someone who had been affected. About 33% admitted that they have fallen for a phishing scam themselves at some point. As shown in figure 7, direct and indirect experiences with phishing scams are fairly common among UMN students, proving the importance of continued cybersecurity education awareness efforts.

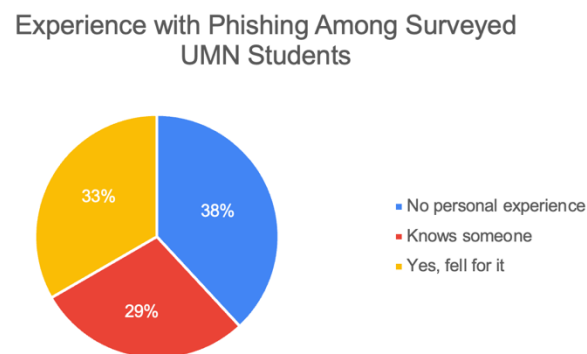


Figure 7 Student experiences with phishing attacks or knowing someone who has been affected.

Two professors from the Carlson School of Management were also interviewed to provide insights. Both expressed concern about the lack of student-targeted training and shared experiences dealing with phishing attempts on campus. One emphasized the effectiveness of simulated phishing tests, while the other highlighted the importance of practical habits like not clicking unknown links. These responses supported the idea that UMN students would highly benefit from being more hands-on and consistent.

In addition to survey results and professor interviews, the University of Minnesota's Information Security Office confirmed that phishing attacks have been increasing among

students and faculty. They reported that common phishing tactics include emails disguised as collaboration documents and survey forms. The office also noted that while mandatory cybersecurity training is required for employees and student workers, general students can voluntarily self-enroll. Furthermore, they talked about ongoing initiatives like providing cybersecurity tips on the university's website and featuring MyPortal reminders during high-risk periods like tax season.

Discussion

The results of this study show that while many UMN students are aware of phishing threats, there are still some gaps that could put them at risk. Nearly 70% of students reported having received a suspicious email or message, but only about 66% felt confident in knowing how to respond to it. Although a majority of students could correctly identify common phishing signs, a portion still felt unsure about what actions to take if faced with a real threat. This pattern suggests that basic awareness alone may not be enough, and that additional training or reminders could help students react more safely and consistently when phishing attempts occur.

While UMN students showed strong knowledge of phishing signs, their confidence and actions when facing real threats align with concerns mentioned in earlier research. Nyasvisvo and Chigada (2023) found that students often recognize phishing tactics but still underestimate how easy it is to fall for a convincing attack. Similarly, Alharbi and Tassaddiq (2021) reported that students might hesitate or make mistakes even when they know the signs of phishing. This matches what was seen in the survey results, where a significant number of students reported feeling only somewhat confident or unsure. Rahartomo (2025) also noted that urgency and legitimate-looking emails are common tricks that continue to catch users off guard, despite rising

awareness. These connections show that knowledge alone may not be enough to fully protect students without practice and strengthening.

During interviews with the professors, they also mention similar concerns about phishing awareness on campus. Professor De Liu mentioned that although UMN occasionally sends phishing awareness emails, there is no consistent or formal training program aimed directly at students. He believed that most students might not be fully aware of how convincing phishing attempts can be, especially when attackers use subtle strategies. Professor James Nord also emphasized the importance of practical habits, such as manually typing in website addresses rather than clicking links. He suggested that the school could implement fake phishing tests, similar to corporate training programs, to give students more hands-on experience. Overall, both professors agreed that additional efforts are needed to keep students more alert and better prepared to handle phishing attacks.

Additionally, a response from the UMN Information Security Office (ISO) confirmed that phishing attempts have been on the rise and that current training efforts are primarily catered to employees and student workers, rather than the general student population. This aligns with the survey findings that while students are aware of phishing threats, gaps in consistent training and education still exist.

Some notable limitations were observed during the study, one is the relatively small sample size. Although 53 students completed the survey, it represents only a small fraction of the entire UMN student population. As a result, the findings may not provide the full insight regarding cybersecurity awareness across all colleges and departments. Additionally, the survey relied on self-report responses, which can introduce bias. Some students may have overestimated their confidence or underreported risky behaviors. Finally, while the University of Minnesota

ISO confirmed an increase in phishing attempts and described general awareness initiatives, their responses were brief. These factors may have limited the depth of comparison between student perceptions and the actual data on phishing threats.

Another growing concern is the rise of artificial intelligence in making phishing attacks more dangerous. As AI technology improves, attackers are able to use it to create phishing messages that are more realistic, harder to detect, and customized to specific targets. Arif, A. Khan, and M. Khan (2024) highlighted that AI-generated cyber threats are becoming more common and sophisticated, which could make traditional phishing training less effective over time. As phishing tactics continuously evolve with new technology, universities like UMN will need to regularly update their cybersecurity education efforts to keep pace with these developments.

Based on the findings of this study, several steps could be taken into consideration in order to improve phishing awareness among UMN students. First, implementing a short and accessible cybersecurity training module could help with gaps in student confidence and behavior. The survey results showed that a majority of students would support such training, especially if it's quick and easy to complete. In addition to that, UMN could run simulated phishing tests to give students practical experience in identifying real threats, a method supported by suggestions from staff interviews. To encourage participation, small incentives such as extra credits can be offered. Ongoing reminders through posters, newsletters, or brief email tips could also keep phishing risks visible without overwhelming students. By combining education with regular engagement, the university could strengthen its students' ability to recognize and respond safely to phishing attacks.

Overall, the results of this study show that while phishing awareness at UMN is fairly strong, there are still areas that need improvement. Students are generally aware of common phishing tactics but may lack the training and practice needed to respond correctly in real situations. As phishing threats continue to evolve with new technologies like AI, it will be important for universities to keep cybersecurity education updated, brief, accessible and engaging. Continued efforts in this area will help students stay better protected against phishing attacks in the future.

Conclusion

This study explored phishing awareness and preparedness among UMN students, highlighting both strengths and areas for improvement. While most students showed an ability to recognize common phishing tactics, a significant number still lacked full confidence in responding to potential attacks. Interviews with professors and feedback from the university's Information Security Office confirmed these gaps and emphasized the need for student-focused, practical security training. As phishing threats continue to evolve, especially with the rise of AI-generated attacks, it will be important for universities to regularly update and improve their education efforts. Implementing short, accessible training modules, running simulated phishing exercises, and providing ongoing reminders could help better prepare students and reduce the risk of phishing-related incidents across the University of Minnesota community.

References

- Alharbi, T., & Tassaddiq, A. (2021). Assessment of cybersecurity awareness among students of Majmaah University. *Big Data and Cognitive Computing*, 5(2), 23. <https://doi.org/10.3390/bdcc5020023>
- Nyasvisvo, B., & Chigada, J. (2023). Phishing attacks: A security challenge for university students studying remotely. *The African Journal of Information Systems*, 15(2), Article 3. <https://digitalcommons.kennesaw.edu/ajis/vol15/iss2/3>
- Rahartomo, A., Ghaleb, A. T. A., & Ghafari, M. (2025). Phishing awareness via game-based learning. *Technische Universität Clausthal* <https://arxiv.org/abs/2501.12077>
- Pincus, E. (2025). Security awareness training in higher education: The ultimate guide. Infosec Institute. <https://www.infosecinstitute.com/resources/security-awareness/ultimate-guide-higher-education/>
- Arif, A., Khan, M. I., & Khan, A. R. (2024). An overview of cyber threats generated by AI. *International Journal of Multidisciplinary Sciences and Arts*, 3(2). <https://www.neliti.com/publications/591914/an-overview-of-cyber-threats-generated-by-i>
- Tran, Tam. "Cybersecurity Awareness at UMN". Survey. 17 April 2025.
- Nord, James & De, Liu. Email Interviews. 21 April 2025.

Survey Questions

1. Have you ever received an email or message that you suspected was a phishing attempt?
2. Do you feel confident in knowing what to do if you get a suspicious email/message?
3. Which of the following signs would you associate with a phishing email/message?
4. What would you do if you received a suspicious email/message?
5. How confident are you in your ability to identify a phishing attempt?
6. Would you support the implementation of a short cybersecurity training module for all UMN students to help prevent phishing attacks and increase awareness?
7. Have you ever fallen for a phishing scam or known someone who has? (Optional)

Professor Interview Questions

1. Have you personally received any phishing emails targeted at UMN faculty or students?
If so, what was it like?
2. Do you think most students in our school are aware of phishing threats?
3. Do you believe the university provides enough cybersecurity education or support for students? Why or why not?
4. In your opinion, would a short phishing awareness module, that may be built into Canvas or orientation, be effective or valuable?

UMN ISO Interview Questions

5. Has the UMN IT department seen an increase in phishing attempts targeting student or faculty in recent years?
6. What are the most common types of phishing tactics reported to IT by students?

7. How does UMN currently educate or alert students about phishing threats (e.g. emails, trainings, orientations, etc.)?
8. In your opinion, would a short phishing awareness module or quiz integrated into Canvas during orientation be an effective way to reduce this issue?
9. Are there any ongoing initiatives or tools the IT department is developing to improve phishing awareness among students?